Auditable Driver-State Annotation from Naturalistic In-Cabin Video

Anonymous Author(s)

ABSTRACT

Naturalistic driving studies increasingly rely on in-cabin video to study driver attention and supervision, yet many datasets are collected without synchronized eye tracking or frame-level driver-state labels. This gap creates a methodological problem for downstream behavioral analysis: researchers need scalable measures of where drivers look and whether their hands remain on the steering wheel, but fully manual video coding does not scale to long-duration, multi-participant recordings. This work-in-progress paper presents an auditable human-in-the-loop workflow for converting ordinary in-cabin video into area-of-interest (AOI) glance categories and hand-on-wheel state variables. The workflow combines reviewed driver-region specification, consensus AOI annotation, participant-specific calibration, frame-level inference, temporal stabilization, hand/wheel detection, hand/wheel state distillation, and window-level behavioral metric extraction. We report preliminary evidence from 15 naturalistic-driving participants, including 301 processed gaze/wheel segments, approximately 3.91 million frame-level outputs, 7,306 coverage-qualified 20 s behavior windows, a five-seed 14-participant leave-one-participant-out AOI experiment, a no-leak few-shot adaptation curve, a temporal stabilization ablation, hand-on-wheel review subsets, a GroundingDINO-to-YOLO state-distillation check, and ONNX deployment checks. The contribution is not a new gaze-estimation backbone or an eye-tracking replacement. Instead, the paper defines a reusable annotation and validation workflow for downstream studies that require transparent, reviewable driver-state features from naturalistic video.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Computer vision tasks*.

KEYWORDS

Naturalistic driving, driver monitoring, glance annotation, human-in-the-loop, area of interest, hand-on-wheel

ACM Reference Format:

Anonymous Author(s). 2026. Auditable Driver-State Annotation from Naturalistic In-Cabin Video. In *AutomotiveUI Work-in-Progress Track, 2026*. ACM, New York, NY, USA, 6 pages.

1 INTRODUCTION

Naturalistic driving studies are valuable because they preserve the conditions under which driver supervision actually occurs: real traffic, real vehicle interfaces, variable lighting, different cabin layouts, and drivers who are not constrained by laboratory tasks. These strengths also make the data difficult to analyze. Many in-cabin video datasets are recorded with ordinary cameras rather than synchronized eye trackers. As a result, they do not directly contain fixations, point-of-regard estimates, or area-of-interest (AOI) glance labels. Before such datasets can support questions about driver monitoring, distraction, takeover readiness, or engagement with driver-assistance systems, researchers must first turn video into behavioral variables.

Manual coding remains the most interpretable way to produce these variables, but it is difficult to scale. Frame-level or clip-level coding requires substantial labor, and reliability checks require preservation of pre-consensus labels or review artifacts. These costs become a bottleneck when a study grows from a small set of clips to hundreds of segments and millions of frames. Fully automated computer vision is appealing, but a black-box classifier is not sufficient for naturalistic driving data. A visually plausible prediction can still be invalid if the driver crop is wrong, if a passenger appears in the selected region, if a face detector fails under glare or occlusion, or if a short prediction flicker is interpreted as a behavioral event.

This paper treats driver-state extraction as a measurement workflow rather than a single model score. Recent models provide useful building blocks: YOLOv8-cls offers a compact train-export-deploy path for image classification [6]; SCRFD provides face evidence that can support driver-region review [4]; and GroundingDINO can localize open-vocabulary objects such as hands and steering wheels [7]. However, these models do not define the study geometry or the behavioral semantics by themselves. In our workflow, humans specify or confirm driver regions, label AOI semantics, review model outputs, and inspect uncertain hand-state cases. Computer vision is used to extend those decisions across long video segments while preserving checkpoints that can be audited.

The resulting measures should be interpreted as AOI-level driver-state features, not continuous gaze estimates or substitutes for eye tracking [3, 5, 8]. The workflow is intended for downstream naturalistic-driving studies that need features such as forward-roadway time, off-path glance episodes, in-vehicle glances, and hands-on/off-wheel ratios. In this WIP, we make four contributions. First, we describe a human-in-the-loop annotation workflow that separates ROI provenance, AOI semantics, temporal stability, hand-state review, and window-level aggregation. Second, we report preliminary evidence for the main checkpoints using reviewable artifacts from 15 participants. Third, we evaluate model and deployment choices so that the workflow can be treated as a measurable annotation method rather than an undocumented preprocessing step. Fourth, we add a state-distillation checkpoint for the slow hand/wheel branch: GroundingDINO is used as a teacher on selected frames, the resulting stable

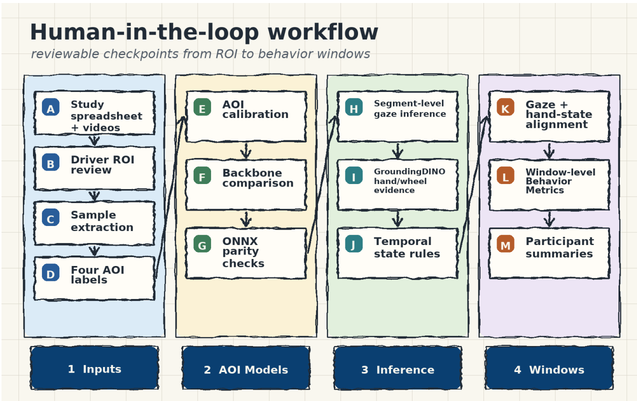

Figure 1: Human-in-the-loop workflow for deriving AOI-level glance and hand-on-wheel variables from naturalistic in-cabin video. The method keeps ROI provenance, frame semantics, temporal stability, hand-state review, and behavior aggregation as separate reviewable stages.

states are converted into a compact student dataset, and the student is evaluated by teacher-state agreement and runtime before any large-scale replacement is claimed.

2 WORKFLOW DESIGN

Figure 1 summarizes the workflow. The input is a set of target videos and analysis segments defined by a study spreadsheet. The workflow resolves local videos, assigns or reviews driver regions, constructs annotation packs, collects human AOI labels, trains participant-specific AOI classifiers, exports deployment models, runs segment-level inference, stabilizes frame-level predictions, estimates hand-on-wheel states, optionally distills GroundingDINO teacher outputs into a lightweight state student, and aggregates the resulting streams into behavioral windows.

2.1 Measurement Targets

The gaze branch assigns each interpretable frame to one of three valid AOI classes. *Forward* denotes attention toward the forward roadway. *In-Car* denotes attention toward in-vehicle targets such as the instrument cluster, center display, mirrors, or controls. *Non-Forward* denotes attention away from both the forward roadway and in-vehicle targets. A fourth label, *Other*, is used for invalid or non-interpretable frames, including wrong ROI, no visible driver, missing or unusable face evidence, non-target person, severe artifact, or insufficient visual information. The primary behavior metrics use the three valid AOI classes, while *Other* is retained for data cleaning, model training, and secondary error analysis.

The hand branch estimates steering-wheel engagement as *ON*, *OFF*, or *UNCERTAIN*. These states are derived from hand and steering-wheel detections, spatial contact evidence, and temporal state rules. For window-level analysis, hand-state samples are aligned with the gaze stream so that researchers can compute both overall and AOI-conditioned hands-on/off ratios.

2.2 Human ROI Specification

ROI selection is treated as part of the measurement method rather than as an invisible preprocessing step. In simple video layouts, a reviewer records the driver ROI directly. In dual-panel or multi-view layouts, face-detection evidence is used to rank candidate driver panels, but the selected panel remains subject to visual confirmation. When automatic assignment is uncertain, the workflow exports grid-overlay reference images and fill-in manifests for manual correction. This design matters because downstream AOI and hand-state predictions are only meaningful if the crop contains the intended driver and the intended body region.

2.3 Human AOI Annotation

After ROI specification, representative frames are sampled across each target video's temporal range and presented for human annotation. Coders assign each ROI crop to *Forward*, *In-Car*, *Non-Forward*, or *Other*. The *Other* class is not treated as a nuisance afterthought. Naturalistic recordings often contain systematic invalid samples caused by occlusion, lighting, camera layout, or non-driver faces. Forcing these samples into a valid AOI would contaminate both model training and downstream behavior summaries.

The annotation process is designed to preserve consensus decisions while also documenting reliability limits. When coder-specific pre-consensus labels are available, chance-corrected agreement can be reported with Cohen's κ [1]; Fleiss' κ is appropriate only when more than two coders independently label the same items [2]. In the current archive, consensus labels and disagreement flags were retained, but complete coder-specific pre-consensus vectors were not retained for every reviewed sample. We therefore treat aggregate pre-consensus agreement as descriptive evidence and report the absence of a reconstructable chance-corrected statistic as a limitation.

2.4 AOI Classification and Participant Adaptation

The operational AOI branch uses a YOLOv8-cls classifier because it provides a compact training and deployment path. Participant-specific calibration starts from a small set of labeled samples and produces a participant-adapted classifier for segment-level inference. At inference time, SCRFD detects faces within the reviewed driver ROI; a layout-specific target-face rule rejects implausible identity jumps and selects the driver face; the selected crop is then classified into the four AOI labels.

Because deployment convenience does not by itself establish measurement validity, the repository also includes a model-comparison workflow that trains YOLOv8-cls, ResNet, EfficientNet, ConvNeXt, and ViT backbones under a common held-out-participant protocol. The comparison is not meant to find a universally best backbone. It tests whether the deployment choice has an irreplaceable accuracy advantage, and whether alternative backbones remain viable when participant generalization or calibration budget is the central concern.

2.5 Temporal Stabilization

Frame-level model outputs are not used directly as behavioral events. Naturalistic video contains short face-detection dropouts, occlusions, lighting transitions, and classifier flicker. The workflow applies presence hysteresis and class debounce rules before computing window-level metrics. Presence hysteresis prevents a brief missed detection from immediately producing an invalid state. Class debounce requires a candidate AOI label to persist before replacing the current stable label. This separation keeps frame semantics and temporal event structure analytically distinct: a classifier can be accurate at the frame level while still producing behaviorally implausible flicker.

2.6 Hand-on-Wheel State Estimation

The hand branch uses the steering-wheel camera view. GroundingDINO detects hands and steering wheels with open-vocabulary prompts. Contact evidence is computed from overlap between detected hand boxes and steering-wheel boxes. The frame-level state machine maps reliable contact evidence to *ON* or *OFF*; unreliable frames retain the previous state for a short grace period before entering *UNCERTAIN*. A temporal majority rule then produces the stable state sequence used for review and metric extraction.

GroundingDINO is useful for bootstrapping because the target objects can be specified by text prompts, but it is too slow to be the only detector for repeated large-scale video processing. The workflow therefore treats GroundingDINO as a teacher detector. Sampled teacher detections can be exported to detection CSV files and reused in two distillation forms: a two-class ROI-crop detector student with *hand* and *steering wheel* labels, or a state-level ROI-crop classifier student supervised by the teacher pipeline's stable *ON*/*OFF*/*UNCERTAIN* outputs. The latter is used as the present acceleration checkpoint because the downstream analysis consumes stable hand-state variables rather than raw boxes.

This branch is intentionally reviewed at the final-state level, after detector confidence, contact rules, and temporal stabilization have acted together. Reviewing only raw boxes would not show whether the derived state sequence is behaviorally plausible. Reviewing only aggregate hands-on percentages would hide whether failures come from the ROI, detector, contact threshold, or temporal rule.

2.7 Window-Level Metrics

The stabilized gaze and hand-state streams are aggregated into fixed analysis windows. Let g_i be the stabilized AOI label at time t_i and let $\mathcal{L} = \{\mathit{Forward}, \mathit{In\text{-}Car}, \mathit{Non\text{-}Forward}\}$ be the valid AOI set. Samples labeled *Other* are excluded from valid-AOI denominators. AOI time percentages are computed with timestamp-based frame-duration weights rather than raw frame counts, so the method can tolerate variation in effective frame rate.

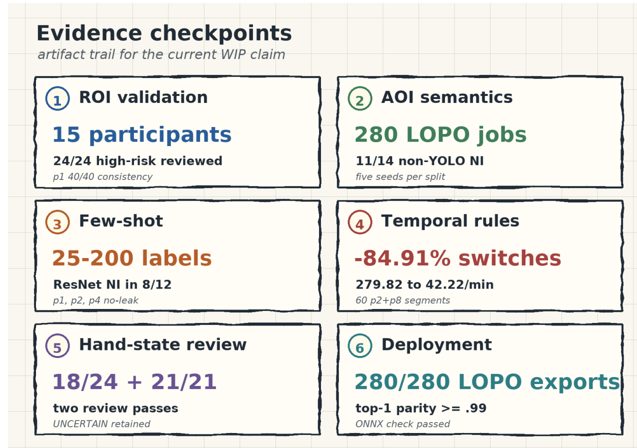

Figure 2: Evidence checkpoints used to evaluate the annotation workflow. Each checkpoint corresponds to a distinct source of possible measurement error.

Glance counts are computed as entries into AOI episodes. Consecutive frames with the same valid AOI label are treated as one episode. Long off-path glances are defined as consecutive frames in either *In-Car* or *Non-Forward*, with event counts reported at 1.6 s and 2.0 s thresholds. Hand-on-wheel ratios are computed overall and conditioned on AOI label after excluding *UNCERTAIN* and unmatched wheel-state samples. Windows that fail the gaze-coverage threshold are retained for quality control but excluded from participant-level behavior summaries.

3 PRELIMINARY EVALUATION

The evaluation focuses on whether the workflow produces reviewable measurement artifacts, not whether it solves driver monitoring in a single end-to-end score. Unless otherwise stated, the evidence below reflects artifacts finalized by May 24, 2026, with processing coverage finalized on May 25, 2026 for the 15-participant analysis set.

3.1 ROI and Processing Coverage

The ROI branch covers 15 participants with human-specified or human-confirmed dual-ROI assignment files and production gaze/wheel ROI manifests. The completed ROI audit covered provenance, resolution, and visual validity of the production ROI set. Its high-risk portion included uncertain p1 dual-ROI rows, p2 low-margin detector-assisted rows, and later layout-specific gaze/wheel checks. All 24 high-risk rows were reviewed to completion and resolved as confirmed, corrected, or no-action cases. The p1 manual/current ROI

Table 1: Processing coverage across the 15-participant analysis set.

Coverage item	Value
Participants in analysis	15
Matched target videos	176
Gaze segments with CSV outputs	301
Wheel segments with CSV outputs	301
Frame-level gaze rows	3,914,416
Frame-level wheel-state rows	3,914,412
Generated 20 s windows	7,327
Coverage-qualified windows	7,306
QC-excluded windows	21

consistency check matched 40/40 rows with zero coordinate changes.

The rebuilt inference plans, gaze maps, wheel maps, and 20 s behavior windows cover all 15 analysis participants. Table 1 summarizes the resulting processing coverage. All 301 planned gaze/wheel segments had frame-level CSV outputs, and all 7,327 generated windows resolved to gaze and wheel files. The final participant-level summaries used 7,306 windows that passed the 0.98 gaze-coverage threshold; 21 windows were retained for quality control but excluded from those summaries.

3.2 AOI Generalization

To evaluate cross-participant generalization, we ran a leave-one-participant-out (LOPO) AOI experiment over 14 held-out participants, five seeds, and four backbones, for 280 training runs. Each split trains on all other participants and freezes the held-out participant as the test set. The primary metric is three-class macro-F1 over *Forward*, *In-Car*, and *Non-Forward*; *Other* remains a secondary data-quality label because the current test sets have limited invalid-sample support.

All 280 LOPO training runs and all 280 prediction jobs completed. Table 2 reports mean held-out performance across the 70 runs per model. ResNet50 achieved the highest mean three-class macro-F1, balanced accuracy, and event-level accuracy. YOLOv8s-cls remains a useful deployment baseline, but these multi-seed point estimates do not show a unique accuracy advantage for YOLOv8s-cls.

At the participant-split level, the best non-YOLO mean exceeded YOLOv8s-cls on 11/14 participants. A bootstrap non-inferiority test with a 3 percentage-point margin found at least one non-YOLO candidate satisfying the margin on 11/14 splits. No non-YOLO model established exact equivalence across the full ± 3 percentage-point band, so the appropriate interpretation is family-best non-inferiority rather than exact equality across all backbones or participants.

Table 2: Mean held-out performance in the five-seed 14-participant LOPO AOI experiment. Each model has 70 held-out runs.

Model	Macro-F1	Bal. acc.	Event acc.
ResNet50	0.596	0.625	0.627
EfficientNet-B0	0.575	0.601	0.616
YOLOv8s-cls	0.574	0.606	0.618
DeiT-Tiny	0.300	0.350	0.399

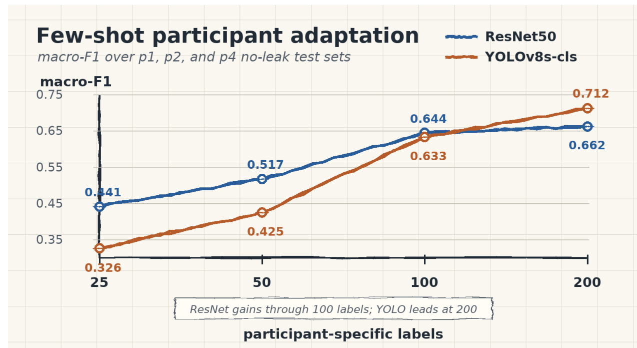

Figure 3: Few-shot no-leak adaptation curve over p1, p2, and p4 (five seeds per participant-budget cell).

3.3 Few-Shot Participant Adaptation

The few-shot experiment tests how much participant-specific labeling is needed when the goal is adaptation rather than zero-label generalization. We ran a no-leak adaptation curve for p1, p2, and p4 at 25, 50, 100, and 200 label budgets, with five seeds per budget and frozen participant-specific test sets. Figure 3 shows rapid ResNet50 gains from 25 to 100 labels, while YOLOv8s-cls catches up and exceeds ResNet50 at 200 labels. The bootstrap non-inferiority test found ResNet50 non-inferior to YOLOv8s-cls in 8/12 participant-budget cells. Thus, label budget and participant adaptation are first-order design variables for this measurement workflow.

3.4 Temporal Stability

We compared raw per-frame AOI predictions with stabilized *Gaze_Class* on a p2+p8 subset covering 60 segments, 643,603 frames, and 436.95 min. Stabilization reduced label switches from 279.82 to 42.22 per minute, an 84.91% reduction. Raw and stable labels differed on 106,418 frames, or 16.54% of the subset. Stabilization also changed behavior-level event structure: 1.6 s off-path episodes rose from 923 to 1,289, and 2.0 s episodes rose from 696 to 1,032. These results show why temporal rules must be evaluated directly rather than treated as cosmetic smoothing.

3.5 Hand-on-Wheel Review

The wheel branch produced 301 segment-level outputs across all 15 analysis participants, totaling 3,914,412 state-classified

Table 3: Hand-on-wheel final-state validation outcomes.

Scope	Reviewed	Agreement	False ON/OFF
Initial manifest	24	18/24	4/2
Corrected second pass	21	21/21	0/0

rows. The uncertainty rate remains high for several participants, including p15 with 75.2% uncertain frames, p17 with 68.6%, and p18 with 70.3%. These values indicate that hand-state estimation is more sensitive to camera layout, occlusion, and steering-wheel evidence than AOI glance classification.

The first agreement summary contained 24 reviewed rows with matched pipeline and human labels. Agreement was 18/24 (75.0%), with four false-ON and two false-OFF cases. During review, we found that one evidence batch initially reused single-panel gaze ROIs as wheel ROIs. We therefore regenerated layout-specific cockpit evidence for the affected participants. The corrected second pass covered seven participants and 21 final states; the reviewer judgment matched all 21 pipeline final states, including 12 binary ON/OFF cases and nine *UNCERTAIN* cases (Table 3). This supports plausibility after ROI-evidence correction, while still leaving independent double-coding as open work.

3.6 Hand/Wheel Distillation Check

We ran a bounded state-distillation check to test whether slow GroundingDINO hand/wheel inference can be amortized through a compact student model. Two p1 GroundingDINO runtime probes, 60 s and 140 s, were converted into 5,000 ROI crops labeled by the teacher pipeline's stable *ON*/*OFF*/*UNCERTAIN* state. A deterministic frame-hash split held out 1,035 validation crops, including all three states. A YOLOv8n-cls student trained from the local YOLOv8n-cls checkpoint with low augmentation reached 0.988 top-1 teacher-state accuracy by Ultralytics validation.

Table 4 summarizes the independent agreement and runtime evidence. Re-running the saved student on the held-out validation manifest produced 1,022/1,035 matching states, or 0.987 agreement with the GroundingDINO teacher states. The confusion counts were 531 $\mathit{OFF}\rightarrow\mathit{OFF}$, 12 $\mathit{ON}\rightarrow\mathit{ON}$, 479 $\mathit{UNCERTAIN}\rightarrow\mathit{UNCERTAIN}$, and 13 total mismatches. After model loading and warm-up, the student processed all 5,000 ROI crops in 6.342 s, or 1.268 ms per crop. The same two GroundingDINO probes previously required 172.752 s, giving a $27.2\times$ loaded-model speedup on the teacher-labeled probe set. This check demonstrates distillation effectiveness without requiring the remaining video segments to be re-inferred.

3.7 Deployment Checks

Deployment checks support the engineering feasibility of the workflow. In the five-seed LOPO matrix, all 280 trained models exported to ONNX and no top-1 parity result fell below 0.99. Table 5 reports a compact CPU benchmark from the holdout protocol, where all 120 exports completed. YOLOv8n/s are smaller and faster than the non-YOLO backbones; ResNet50 is the fastest non-YOLO option but is much larger on disk. These results support YOLOv8-cls as a convenient deployment path even though it is not uniquely accurate in the held-out AOI experiments.

Table 4: GroundingDINO-to-YOLO hand-state distillation check.

Check	Result
Teacher-labeled subset	2 p1 probes; 5,000 ROI crops; 3 stable states
Split and student	3,965 train / 1,035 validation; YOLOv8n-cls, 1.44M params
Teacher-state agreement	0.987 validation agreement; 1,022/1,035 states match
Runtime probe	6.342 s for 5,000 crops; $27.2\times$ vs. GroundingDINO teacher wall time

Table 5: Compact ONNX CPU deployment benchmark from the holdout protocol.

Model	MB	B1 p50 ms	B1 img/s	B32 img/s
YOLOv8n-cls	5.5	4.85	202.2	1621.2
YOLOv8s-cls	19.4	5.26	183.7	903.6
ResNet50	89.6	7.20	139.1	525.7
DeiT-Tiny	21.2	17.19	56.7	291.1
EfficientNet-B0	15.3	18.63	53.4	288.3
ConvNeXt-Tiny	106.2	18.65	51.8	114.1
EfficientNet-B3	40.8	33.69	29.4	157.5

4 DISCUSSION

The workflow addresses a recurring measurement problem in naturalistic-driving research: studies often need driver-state variables from long in-cabin videos, but the available data do not contain eye-tracking streams or hand-state labels. The central design choice is to keep the annotation process auditable. ROI selection, AOI semantics, participant calibration, temporal stability, hand-state review, and window aggregation are separate stages, so errors can be localized and corrected rather than hidden inside an aggregate model score.

The preliminary evidence also clarifies the role of the computer-vision backbones. YOLOv8-cls is attractive because it is compact, easy to train, and straightforward to export. The AOI experiments, however, suggest that deployment convenience should not be confused with unique accuracy. ResNet50 and EfficientNet-B0 remain competitive under participant generalization and few-shot adaptation. For hand-on-wheel estimation, GroundingDINO is useful for open-vocabulary bootstrapping and review, but its computational cost and uncertainty rates suggest that a smaller task-specific student may be preferable for large-scale deployment. The state-distillation check shows one practical route: spend the GroundingDINO cost once on selected teacher frames, then use a compact student classifier to reproduce stable hand-state outputs for faster future inference.

For downstream studies, the practical value of this method is that it turns video into documented behavioral variables without requiring each study to re-derive the full annotation pipeline. A later analysis can cite this workflow as the source of AOI glance and hand-on-wheel feature construction, while focusing its own claims on the behavioral or psychological question under study. That citation relationship requires the present paper to remain method-focused: it defines the measurement process, evidence checks, and limitations, but it does not claim the substantive results of any downstream trust or automation-use analysis.

5 LIMITATIONS AND OPEN WORK

Several limitations remain. First, the workflow depends on participant labels and is not a zero-label generalization system. Second, *Other* is important for robustness but remains difficult to evaluate when frozen test sets contain few invalid samples. Third, ROI transcription or layout-selection errors can dominate downstream results; the current production manifests have completed visual ROI review, but new layouts, videos, or manifest edits would require repeated review. Fourth, the corrected hand-on-wheel second pass is single-reviewer evidence and should be expanded into an independently double-coded validation set. Fifth, the current distillation check is p1-derived and uses a frame-hash validation split, so the student should be retrained and stress-tested on broader multi-participant teacher samples before replacing GroundingDINO in production. Finally, deployment benchmarks should be repeated on the target runtime hardware before strong latency claims are made.

6 CONCLUSION

This WIP presents a human-in-the-loop workflow for extracting AOI-level glance and hand-on-wheel variables from naturalistic in-cabin video. The workflow combines human ROI review, AOI annotation, participant-specific calibration, segment-level inference, temporal stabilization, hand-on-wheel review, state distillation, and window-level metric extraction. Preliminary evidence from 15 participants shows that the pipeline can process millions of frames while preserving checkpoints for ROI validity, AOI generalization, adaptation cost, temporal stability, hand-state plausibility, student acceleration, and deployment feasibility. The resulting contribution is a transparent measurement method for downstream naturalistic-driving studies, not a replacement for eye tracking or a standalone theory of driver behavior.

REFERENCES

[1] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20, 1 (Apr 1960), 37–46. https://doi.org/10.1177/001316446002000104

[2] Joseph L. Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological Bulletin* 76, 5 (Nov 1971), 378–382. https://doi.org/10.1037/h0031619

[3] Lex Fridman, Philipp Langhans, Joonbum Lee, and Bryan Reimer. 2016. Driver gaze region estimation without use of eye movement. *IEEE Intelligent Systems* 31, 3 (May 2016), 49–56. https://doi.org/10.1109/MIS.2016.47

[4] Jia Guo, Jiankang Deng, Alexandros Lattas, and Stefanos Zafeiriou. 2021. Sample and computation redistribution for efficient face detection. *arXiv preprint arXiv:2105.04714* (May 2021). arXiv:2105.04714 [cs.CV] https://arxiv.org/abs/2105.04714

[5] International Organization for Standardization. 2020. *Road vehicles – measurement and analysis of driver visual behaviour with respect to transport information and control systems.* International Standard ISO 15007:2020. International Organization for Standardization. https://www.iso.org/standard/63220.html

[6] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. 2023. Ultralytics YOLOv8. Software. https://github.com/ultralytics/ultralytics

[7] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. 2024. Grounding DINO: marrying DINO with grounded pre-training for open-set object detection. In *Computer Vision – ECCV 2024: 18th European Conference, Milan, Italy, September 29–October 4, 2024, Proceedings, Part XLVII*, Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol (Eds.). Lecture Notes in Computer Science, Vol. 15105. Springer Nature Switzerland, Cham, 38–55. https://doi.org/10.1007/978-3-031-72970-6_3

[8] SAE International. 2023. *Definitions and experimental measures related to the specification of driver visual behavior using video-based techniques.* SAE Recommended Practice J2396_202309. SAE International. https://doi.org/10.4271/J2396_202309